

DRY-RUN GUIDE | magic.py

Source: Python\3rd_Belt for practice\Matrix\magic.py

CORE BEHAVIOR

Checks that every row, column, and both diagonals have the same sum.

REPRESENTATIVE INPUT

Lo Shu square: '8 1 6' / '3 5 7' / '4 9 2'.

STEP TRACE

Target row sum=15; every row, column, and diagonal sums to 15.

EXPECTED OUTPUT / RETURN

'True'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt for practice\Matrix\magic.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Check whether a square matrix is a magic square.
002 | # Add your solution here.
003 |
004 | def is_magic_square(matrix, N):
005 |     target_sum = sum(matrix[0])
006 |
007 |     for row in matrix:
008 |         if sum(row) != target_sum:
009 |             return False
010 |
011 |     for col in range(N):
012 |         col_sum = sum(matrix[row][col] for row in range(N))
013 |         if col_sum != target_sum:
014 |             return False
015 |
016 |     diag1_sum = sum(matrix[i][i] for i in range(N))
017 |     if diag1_sum != target_sum:
018 |         return False
019 |
020 |     diag2_sum = sum(matrix[i][N - 1 - i] for i in range(N))
021 |     if diag2_sum != target_sum:
022 |         return False
023 |
024 |     return True
025 |
026 | N = int(input())
027 | matrix = [list(map(int, input().split())) for _ in range(N)]
028 |
029 | print(is_magic_square(matrix, N))
030 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.